

TABLE DESIGN – CAMPUSPULSE

1. Users Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	name	String (100)	Required	Full name of the user
3	email	String (100)	Unique, Required	Email address used for login
4	password	String (255)	Required	Bcrypt-hashed password
5	role	String (20)	Required, Indexed	User role: Student, Recruiter, TPO
6	phone	String (15)	Optional	Contact number
7	isActive	Boolean	Required	Account activation status
8	createdAt	Date	Required	Account creation timestamp
9	updatedAt	Date	Required	Last update timestamp

2. Students Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	userId	ObjectId (24 chars)	Foreign Key (Users), Unique, Required	Reference to Users._id
3	rollNumber	String (20)	Unique, Required, Indexed	College roll number
4	cgpa	Number (4,2)	Required, Indexed	Current CGPA (out of 10)
5	branch	String (50)	Required, Indexed	Department/Branch
6	passoutYear	Number (4)	Required	Year of graduation
7	activeBacklogs	Number (2)	Required	Number of active backlogs
8	resumePath	String (24)	Optional	GridFS fs.files._id (file ID)
9	isProfileComplete	Boolean	Required	Profile completion status

3. Recruiters Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	userId	ObjectId (24 chars)	Foreign Key (Users), Unique, Required	Reference to Users._id
3	companyName	String (100)	Required, Indexed	Company/organization name
4	gstNumber	String (15)	Optional	GST identification number
5	website	String (100)	Optional	Company website URL
6	officialEmail	String (100)	Unique, Required	Company's official email
7	companyLogo	String (24)	Optional	GridFS file ID for company logo
8	isApproved	Boolean	Required, Indexed	TPO approval status
9	verifiedAt	Date	Optional	Date of verification

4. JobDrives Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	recruiterId	ObjectId (24 chars)	Foreign Key (Recruiters), Required, Indexed	Reference to Recruiters._id
3	title	String (100)	Required	Job title/role name
4	description	String (Large)	Required	Job description
5	ctc	Number (10)	Required	Annual Cost to Company (INR)
6	minCGPA	Number (4,2)	Required, Indexed	Minimum CGPA required
7	eligibleBranches	Array of Strings	Required, Indexed	List of eligible branches
8	maxBacklogs	Number (2)	Required	Maximum backlogs allowed
9	applicationDeadline	Date	Required	Last date to apply
10	hasAptitudeTest	Boolean	Required	Aptitude Test included?
11	hasGD	Boolean	Required	Group Discussion included?
12	status	String (15)	Required, Indexed	Draft, Open, Closed
13	createdAt	Date	Required	Drive creation timestamp
14	updatedAt	Date	Required	Last update timestamp

5. Applications Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	studentId	ObjectId (24 chars)	Foreign Key (Students), Required, Compound Index	Reference to Students._id
3	driveId	ObjectId (24 chars)	Foreign Key (JobDrives), Required, Compound Index	Reference to JobDrives._id
4	appliedDate	Date	Required	Application submission date
5	status	String (30)	Required, Indexed	Multi-stage status (see list below)
6	aiMatchScore	Number (4,3)	Optional	AI-generated match score (0-1)

Status Values: Applied, Under Review, Aptitude Scheduled, Aptitude Completed, GD Scheduled, GD Completed, Interview Scheduled, Interview Completed, Selected, Rejected, Waitlisted, Offer Sent, Offer Accepted, Offer Declined, Placed

Compound Index: { driveId: 1, status: 1, aiMatchScore: -1 }

Embedded: aptitude (Object)

Sl. No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	aptitude.status	String (20)	Optional	Scheduled, Completed, Passed, Failed, Not Applicable
2	aptitude.score	Number (3)	Optional	Test score
3	aptitude.markedBy	ObjectId (24 chars)	Foreign Key (Recruiters)	Recruiter who marked the result
4	aptitude.markedAt	Date	Optional	Date of result declaration

5.3 Embedded: gd (Object)

Sl. No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	gd.status	String (20)	Optional	Scheduled, Completed, Shortlisted, Rejected, Not Applicable
2	gd.markedBy	ObjectId (24 chars)	Foreign Key (Recruiters)	Recruiter who marked the result
3	gd.markedAt	Date	Optional	Date of result declaration

5.4 Embedded: interview (Object)

Sl. No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	interview.status	String (15)	Optional	Scheduled, Completed , Pending
2	interview.result	String (15)	Optional	Selected, Rejected, Waitlisted, Pending
3	interview.feedback	String (500)	Optional	Recruiter's feedback
4	interview.markedBy	ObjectId (24 chars)	Foreign Key (Recruiters)	Recruiter who marked the result
5	interview.markedAt	Date	Optional	Date of result declaration

5.5 Embedded: xai (Object – Explainable AI Feedback)

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	xai.matchScore	Number (4,3)	Optional	Overall match score
2	xai.positiveSentences	Array of Strings	Optional	Sentences that helped the score
3	xai.negativeSentences	Array of Strings	Optional	Sentences that hurt the score
4	xai.skillGaps	Array of Strings	Optional	Skills missing from the resume
5	xai.strongSkills	Array of Strings	Optional	Strong skills identified

5.6 Embedded: offer (Object – Offer Letter)

Sl. No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	offer.status	String (15)	Optional	Sent, Viewed, Downloaded, Accepted, Declined, Expired
2	offer.fileId	String (24)	Optional	GridFS fs.files._id (PDF file ID)
3	offer.fileName	String (100)	Optional	Original file name
4	offer.uploadedDate	Date	Optional	Date of upload
5	offer.expiryDate	Date	Optional	Offer validity expiry date
6	offer.viewedAt	Date	Optional	Date student viewed
7	offer.downloadedAt	Date	Optional	Date student downloaded
8	offer.acceptedAt	Date	Optional	Date of acceptance
9	offer.declinedAt	Date	Optional	Date of declination
10	offer.declineReason	String (255)	Optional	Reason for declining

6. Schedules Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	studentId	ObjectId (24 chars)	Foreign Key (Students), Required, Composite Index	Reference to Students._id
3	recruiterId	NEW ObjectId (24 chars)	Foreign Key (Recruiters), Required, Indexed	Reference to Recruiters._id (for calendar views)
4	driveId	ObjectId (24 chars)	Foreign Key (JobDrives), Required	Reference to JobDrives._id
5	applicationId	ObjectId (24 chars)	Foreign Key (Applications), Required	Reference to Applications._id
6	eventType	String (15)	Required	Aptitude, GD, Interview
7	date	Date	Required, Composite Index	Event date
8	timeSlot	String (30)	Required, Composite Index	Start - End time (e.g., 10:00 AM - 11:00 AM)
9	location	String (255)	Optional	Venue or online meeting link
10	status	String (15)	Required	Scheduled, Completed, Cancelled
11	reminderSent	Boolean	Required	24-hour email reminder sent?
12	createdAt	Date	Required	Schedule creation timestamp

Composite Index: { studentId: 1, date: 1, timeSlot: 1 } (For conflict detection)

7. RecruiterTrustScore Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	recruiterId	ObjectId (24 chars)	Foreign Key (Recruiters), Unique, Required	Reference to Recruiters._id
3	domain	String (100)	Required	Company website domain
4	whoisData	Object (JSON)	Optional	Raw WHOIS lookup data
5	mcaData	Object (JSON)	Optional	MCA registration data
6	trustScore	Number (3)	Required	Calculated trust score (0-100)
7	verifiedAt	Date	Required	Date of verification

8. EmailLogs Collection

Sl.No	Field Name	Data Type (Size)	Key Constraints	Description Of the Field
1	_id	ObjectId (24 chars)	Primary Key	Auto-generated unique identifier
2	recipientEmail	String (100)	Required, Indexed	Recipient's email address
3	recipientId	ObjectId (24 chars)	Foreign Key (Users)	Reference to Users._id
4	emailType	String (20)	Required	Activation, Reminder, Result, Offer
5	subject	String (200)	Required	Email subject line
6	sentAt	Date	Required	Date and time of sending
7	status	String (15)	Required, Indexed	Sent, Failed, Opened, Bounced
8	errorMessage	String (500)	Optional	Error details (if failed)